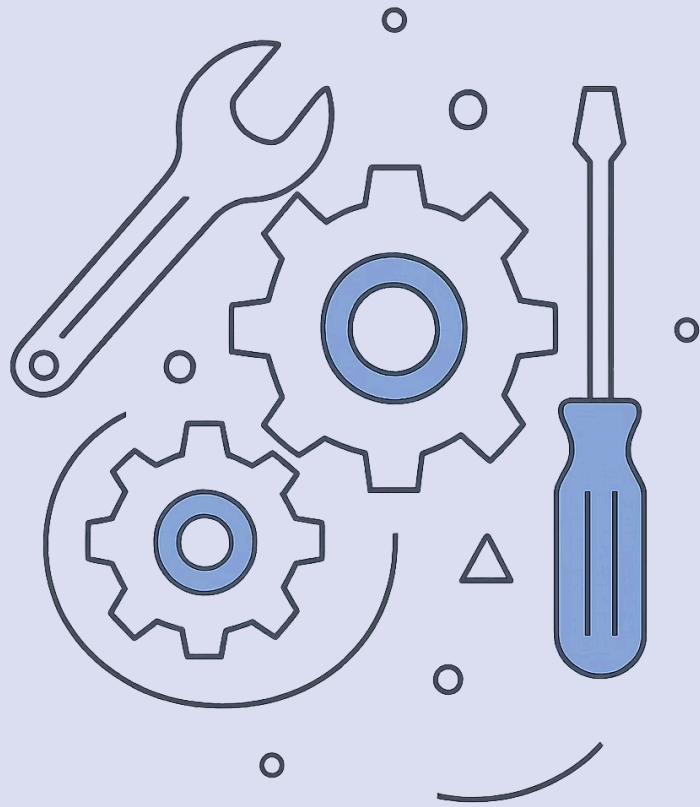




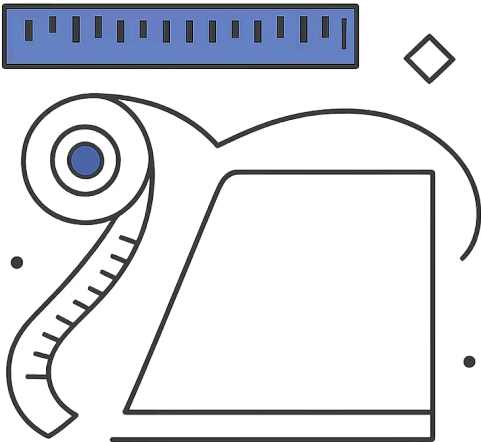
Part 4: Practical Issues

Real-world problems in Multiple Linear Regression.

Why Scaling Matters

Features may have very different ranges, making direct coefficient comparison difficult.

Feature	Range	Unit
Study Hours	1-10	hours
Attendance	40-100	%
Previous Score	0-100	marks
Sleep Hours	4-9	hours



Feature Scaling Methods

Feature scaling converts features into a common scale for fair comparison.

Min-Max Scaling

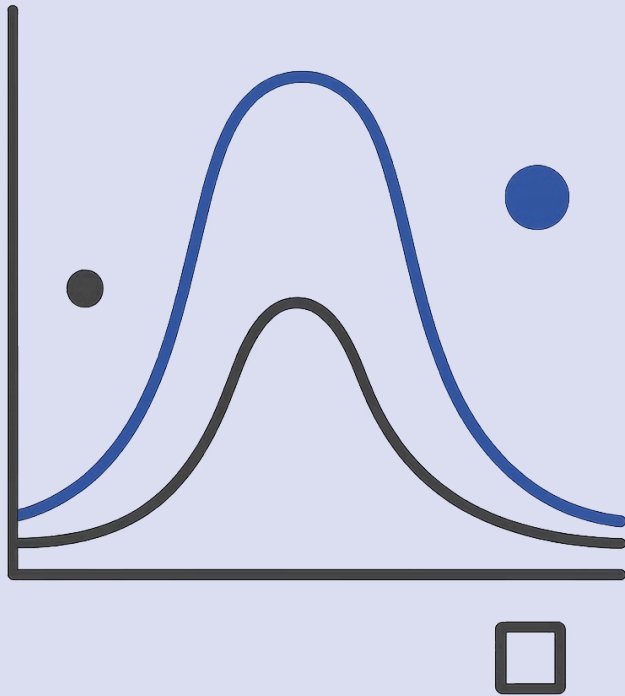
Formula: $(x - \min) / (\max - \min)$

Rescales values to a 0–1 range.

Standardisation

Formula: $(x - \text{mean}) / \text{standard deviation}$

Centres data around mean = 0, SD = 1.



Standardisation

Formula: $z = (x - \text{mean}) / \text{standard deviation}$

Mean $\rightarrow 0$

Data is centred at zero after transformation.

SD $\rightarrow 1$

Spread is normalised to one unit.

z-score meaning

Tells how far a value is from the average.

Coefficients Before Scaling

Raw coefficients are not directly comparable because units differ.

Feature	Coefficient	Unit Meaning
Study Hours	2.5	Per 1 hour
Attendance	0.2	Per 1 percentage point
Previous Score	0.5	Per 1 mark
Sleep Hours	1.5	Per 1 hour

Coefficients After Scaling

Scaled coefficients allow meaningful comparison of feature importance.

Feature	Scaled Coefficient
Previous Score	6.5
Study Hours	4.8
Attendance	3.2
Sleep Hours	1.4

Each coefficient now represents the effect of a 1 standard deviation increase.

Data Leakage Warning

❌ Wrong approach: Scale the full dataset first, then split into train and test.

✅ Correct approach: Split first. Learn scaling values from training data only. Apply the same scaling to test data.

Test data must remain unseen during any preprocessing step.



Target Leakage — A Dangerous Mistake

Target leakage means a feature secretly contains information about the answer. The model looks excellent in training but fails in real life.

Feature	Problem
Final grade status	Already depends on final score
Certificate eligibility	Decided after score
Result published flag	Future information
Rank after exam	Contains target information

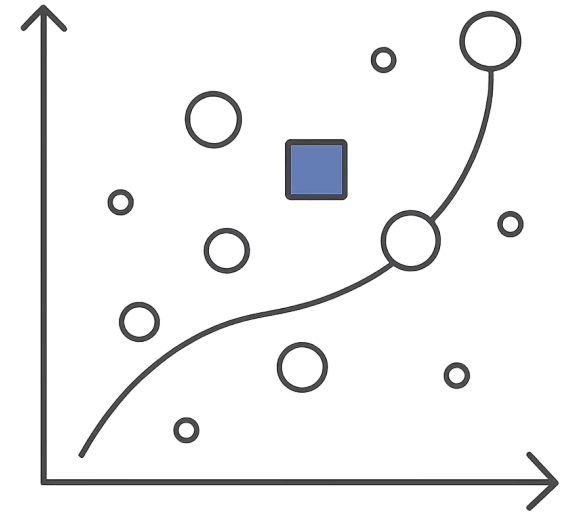
Residual Analysis — Check the Errors

Residual = Actual – Predicted

Good residual behaviour:

- Errors look random
- No clear pattern
- No extreme outliers
- Error spread is roughly stable

 If residuals show a pattern, the model may be missing something.



Residual Pattern Warning

Residuals help diagnose model weakness.

Residual Pattern	Possible Meaning
Curved pattern	Relationship may not be linear
Increasing spread	Error variance is changing
Many large errors	Outliers or missing features
Group-wise errors	Category or segment effect missing

Multicollinearity — When Features Duplicate Information

Multicollinearity means two or more input features are strongly related to each other.

Feature 1	Feature 2
Study Hours	Practice Tests
House Size	Number of Rooms
Experience	Age
Ad Spend	Impressions

Why Multicollinearity Is a Problem

The Model's Dilemma

If two features say almost the same thing, the model gets confused about how much credit to assign each one.

Prediction vs. Interpretation

Prediction may still be good, but coefficient interpretation becomes unreliable.

Correlation

Correlation shows how two features move together.

Correlation	Meaning
+1.0	Perfect positive relationship
+0.9	Very strong positive relationship
0.0	No linear relationship
-0.9	Very strong negative relationship
-1.0	Perfect negative relationship

Multicollinearity Example

Student	Study Hours	Practice Tests
A	2	1
B	3	2
C	4	3
D	5	4
E	6	5

Study Hours and Practice Tests move together perfectly — they may duplicate information, causing multicollinearity.



The model cannot reliably separate the effect of each feature.

Coefficient Confusion

When features are strongly correlated, the model may produce strange coefficients:

Study Hours

Coefficient = -1.5

Does studying reduce score? Probably not.

Practice Tests

Coefficient = $+6.0$

Inflated to compensate for the correlated feature.

This happens because the two features are strongly correlated — coefficients become unreliable.

VIF — Variance Inflation Factor

VIF measures how much one feature can be explained by other features.

Formula: $VIF = 1 / (1 - R^2)$

VIF	Meaning
1	No multicollinearity
1-5	Usually acceptable
5-10	Warning zone
Above 10	Serious issue

Fixing Multicollinearity



Remove a Feature

Keep the more useful of two correlated features.



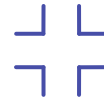
Regularisation

Ridge regression can help manage multicollinearity.



Combine Features

Create a new meaningful composite score.

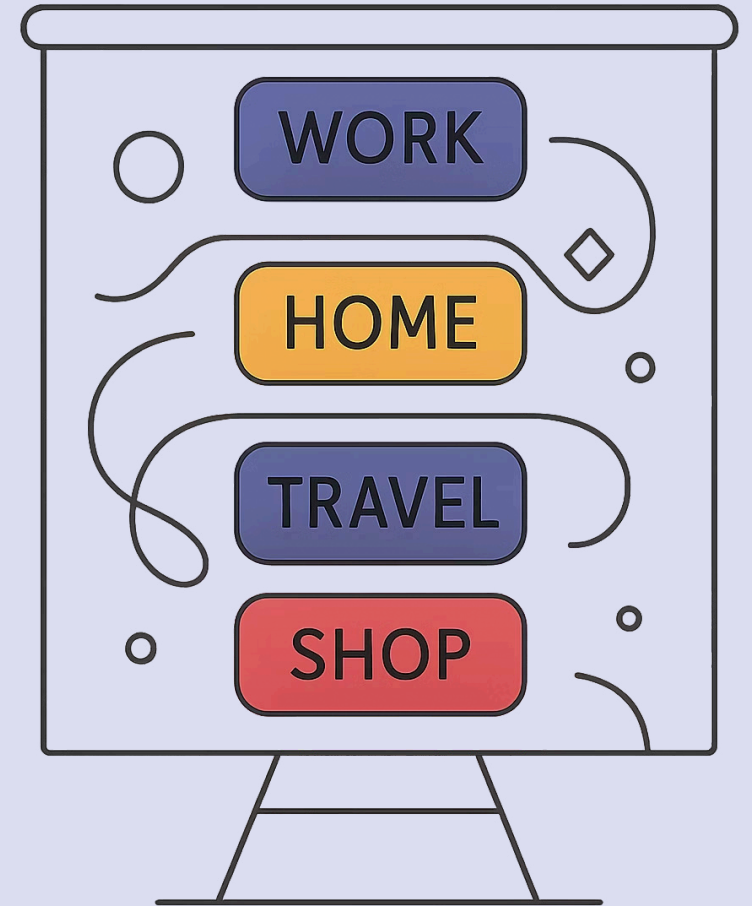


Use PCA

Advanced feature compression to reduce overlap.

Part 5: Categorical Features and Interactions

Handling real-world data types in Multiple Linear Regression.



Categorical Features

Real datasets often contain text columns. Linear Regression needs numbers, so categories must be encoded.

Feature	Categories
School Type	Government, Private
Learning Mode	Online, Offline
City	Chennai, Hyderabad, Bengaluru
Department	CSE, ECE, Mechanical

Nominal vs Ordinal

Nominal

No natural order — e.g. City, School Type

Use One-Hot Encoding

Ordinal

Has a natural order — e.g. Low, Medium, High

Use ordered encoding



Do not assign random numbers to unordered categories — this creates a false order.

One-Hot Encoding

Original:

Student	School Type
A	Government
B	Private
C	Private
D	Government

Encoded:

Student	School Type_Private
A	0
B	1
C	1
D	0

Government is the reference category.

The Dummy Variable Trap

With two categories (Government / Private), creating both columns gives duplicate information — one already tells the other.

Government	Private
1	0
0	1

- ✓ Solution: Use $k - 1$ dummy columns. Drop one category as the reference.



Dummy Coefficient Interpretation

Model: $\text{Score} = b_0 + b_1(\text{Study Hours}) + b_2(\text{School Type_Private})$

School Type_Private = 1

Student attends a Private school.

School Type_Private = 0

Student attends a Government school (reference).

Meaning of b_2

Effect of Private compared to Government, keeping other features fixed.

“Dummy coefficients are always interpreted compared to reference category.”

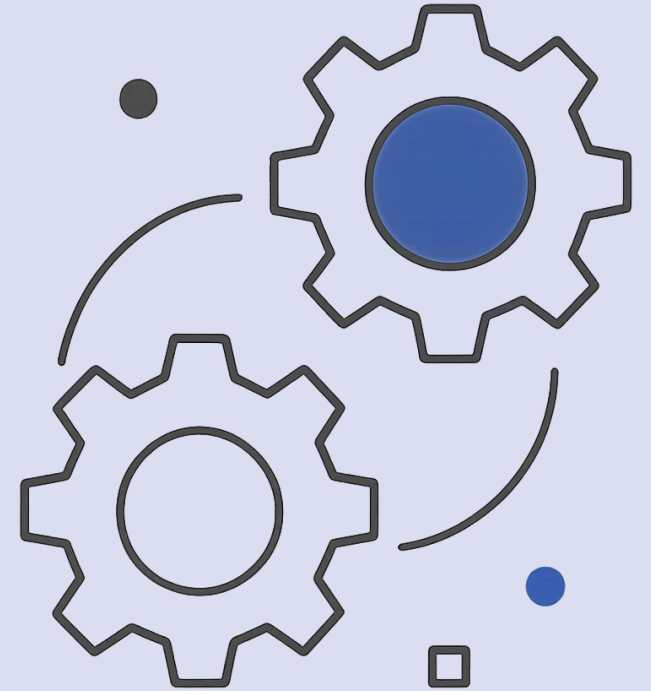
Interaction Terms

Sometimes one feature's effect depends on another.

Example: Study hours may help more when previous score is already strong.

Interaction term: Study Hours $\times$ Previous Score

$$\text{Score} = b_0 + b_1(\text{Study Hours}) + b_2(\text{Previous Score}) + b_3(\text{Study Hours} \times \text{Previous Score})$$



Interaction Example

Study hours may help more when previous score is already strong.

Student	Study Hours	Previous Score	Interaction
A	2	40	80
B	3	45	135
C	4	50	200
D	5	60	300

Interaction = Study Hours × Previous Score

Meaning of the Interaction Coefficient

Positive Coefficient

The two features strengthen each other. Study hours become more useful when previous score is higher.

Negative Coefficient

One feature reduces the effect of another. Study hours become less effective when stress is high.

Interaction Prediction Example

Model:

$$\text{Score} = 10 + 2(\text{Study Hours}) + 0.5(\text{Previous Score}) + 0.02(\text{Study} \times \text{Previous})$$

Student: Study Hours = 4, Previous Score = 70

$$\text{Interaction} = 4 \times 70 = 280$$

✓ $\text{Score} = 10 + 8 + 35 + 5.6$

$$\text{Score} = 58.6$$

Interaction Interpretation Rule

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3(x_1x_2)$$

Effect of x_1

$$= b_1 + b_3x_2$$

Key Insight

The effect of x_1 changes depending on x_2 . That is the mathematical meaning of interaction.

Assumptions of Multiple Linear Regression

Assumption	Meaning
Linear relationship	Features combine linearly
Numeric target	Output is a number
Independent errors	Errors should not depend on each other
Low multicollinearity	Features should not duplicate too much
Controlled outliers	Extreme points should be checked
Constant error spread	Error variance should be stable

Common Mistakes

- Too many features
Adding features blindly can cause overfitting.
- Ignoring multicollinearity
Coefficients become unstable.
- Scaling before splitting
Causes data leakage.
- Encoding with random numbers
Creates a false order for nominal categories.
- Leaked target information
Gives fake performance metrics.

When to Use Multiple Linear Regression

Use it when:

- Target is numeric
- Multiple features influence the target
- Relationship is mostly linear
- Interpretability is important

Domain	Prediction
Education	Final score
Real estate	House price
HR	Salary
Marketing	Sales
Finance	Loan amount

Final Summary

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \dots$$

Prediction = base value + contribution of each feature

Training goal: Find coefficients that minimise error.

- Coefficients
- Evaluation
- Train-Test Split
- Scaling
- Multicollinearity
- One-Hot Encoding
- Interaction Terms